



Contents lists available at ScienceDirect

Research in Autism Spectrum Disorders

Journal homepage: <http://ees.elsevier.com/RASD/default.asp>



Evaluating the effectiveness of two commonly used discrete trial procedures for teaching receptive discrimination to young children with autism spectrum disorders

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ARTICLE INFO

Article history:

Received 16 December 2008

Accepted 16 December 2008

Keywords:

Autism

Discrimination training

Discrete trial

Receptive language

ABSTRACT

Discrete trial teaching procedures have been demonstrated to be effective in teaching a variety of important skills for children with autism spectrum disorders (ASD). Although all discrete trial programs are based in the principles of applied behavior analysis, some variability exists between programs with regards to the precise teaching procedures used. One notable procedural discrepancy involves teaching receptive discriminations. We compared the effectiveness of two commonly used procedures to teach receptive discriminations for three young children with ASD. One procedure progressed from introducing novel target stimuli in isolation (i.e., with no distracter present) to a conditional discrimination training phase (i.e., with distracter). A second procedure involved teaching discriminations exclusively within the context of conditional discriminations (i.e., using distracter stimuli). A within-subject comparison of procedures revealed mixed results and a 1-month follow-up revealed no differences between teaching procedures.

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Young children with autism spectrum disorders (ASD) benefit from early and intensive intervention programs based on the principles of applied behavior analysis (ABA) (McEachin, Smith, & Lovaas, 1993; Smith, 1999). Several comprehensive ABA programs have been developed to address the learning needs of young children with ASD (Arick, Loos, Falco, & Krug, 2005; Leaf & McEachin,

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1999; Lovaas, 1981, 2003; Maurice, Green, & Luce, 1996). These programs are similar in that they are based on operant learning principles and target similar skills for instruction (e.g., expressive and receptive language, imitation, discrimination, socialization). However, the specific procedures used to teach skills sometimes differ among programs. One common procedural discrepancy is the method used to teach receptive discriminations. Receptive discrimination programs (matching to sample) are used to teach “receptive language” or the understanding of what is being said by another person (Green, 2001). Receptive discriminations typically involve a child selecting an item (or picture of an item) from an array of items (or pictures) that corresponds to stated label. For example, when asked to “point to the car,” the child makes a receptive discrimination by pointing to car, but not to other items present.

Currently, two procedures have been described to teach receptive discriminations. One commonly described procedure recommends teaching novel receptive discriminations in a progression from introducing novel target stimuli in isolation (i.e., no distracter present) to a conditional discrimination training phase (i.e., with distracter) (e.g., Arick et al., 2005; Lovaas, 1981, 2003; Maurice et al., 1996). For example, when using this procedure to introduce “car,” the car is presented alone. The child is taught to “point to the car” when the car is the only item available to select. Once the child has mastered pointing to the car (based on a predetermined mastery criterion), then a non-target distracter stimulus is introduced (conditional discrimination). In the conditional discrimination training phase, the car is selected from an array of multiple stimuli (e.g., picture of a car and picture of a hat).

In an alternative procedure, receptive discriminations are taught exclusively in the context of conditional discriminations that include distracter stimuli, thus eliminating the initial simple discrimination (no distracter) training procedure (e.g., Leaf & McEachin, 1999). In other words, novel receptive stimuli are always presented in the context of at least one distracter stimulus. When introducing “point to the car,” a picture of a car and at least one alternative distracter picture (e.g., picture of a hat) are available from the first training session.

This discrepancy in teaching procedures persists among current practitioners as well. Carr, Love, Almason, and Petursdottir (2006) surveyed interventionists providing ABA programs to young children with autism. Of those responding, 37% reported progressing from a simple discrimination training phase to a conditional discrimination training phase for teaching receptive discriminations. Thirty two percent (32%) reported teaching receptive discriminations exclusively within the context of conditional discriminations. The remaining respondents reported using some other unspecified method. Despite the prevalence of both procedures in ABA programs as well as current practice, there is no existing applied literature comparing the efficacy of one procedure over the other.

In a related basic research study, Terrace (1963) evaluated how the introduction of distracter stimuli (S-) early or late during discrimination training effected how readily discriminations were acquired by pigeons. When the distracter stimuli (S-) were introduced early during training, performance on a discrimination task was superior (fewer errors or responses to the S- stimulus) to presentation of the distracter stimuli later in training. These results suggest that discrimination training that progresses from a simple discrimination (no distracter) training to the conditional discrimination (distracter phase) may, in fact, be less effective than procedures which introduce distracter items earlier.

The purpose of this study was to compare the rate of learning receptive discriminations when utilizing these two common teaching procedures.

1. Method

1.1. Participants and setting

Participants were three children with community diagnoses of ASD. Brad was a 24-month-old, non-verbal male. Julie was a 24-month-old female who uses a few single words to communicate. Sal was a 48-month-old male who uses a few single words to communicate. Brad and Sal received research diagnoses of autism as part of other research projects. For Brad and Julie, all sessions were conducted within the context of their pre-school classroom in an area of the room designed for 1:1

teacher-led instruction. For Sal, all sessions were conducted in a clinic-based setting. Participants were seated across from the therapist at a small table. Therapists were graduate and undergraduate students trained in discrete trials methods and supervised by doctoral level, board certified behavior analysts.

1.2. Materials and teaching sessions

Participants were taught receptive discrimination of picture cards using 1:1 discrete trial training (Lovaas, 1981). Physical guidance was used during all procedures to guide participants to the correct response. Reinforcement was provided on a variable ratio schedule of reinforcement (VR 3), with the exception of the pre-treatment assessment and 1-month follow-up, where no reinforcement was provided. Preferred items to be used as reinforcers were identified for each participants upon entering the program and then again prior to each session using a free operant preference assessment. All conditions used comparably difficult exemplars for identification. Table 1 shows the two target stimuli taught for each training set for individual participants.

1.3. Design

The design for the study was a within-subject, adapted alternating treatment design, whereby each participant was taught to receptively identify pictures using two different discrimination training procedures (described further below). The presentation of teaching procedures was counterbalanced across participants to rule out a possible sequence effect. After each initial discrimination training procedure, we further assessed receptive discriminations by pairing two previously taught picture cards for subsequent discrimination training.

1.4. Data collection and inter-observer agreement

Data collectors recorded the occurrence of correct or prompted response for each ten-trial session. Two observers independently collected data for 35% of all sessions. Inter-observer agreement was calculated using the point-by-point agreement ratio and averaged 99.5%.

1.5. Pre-treatment assessment

Prior to beginning each training phase, a pre-treatment assessment was conducted to identify receptive labels that were unknown to each participant. To do this, two picture cards were simultaneously presented and the participant was asked to receptively identify (e.g., point to) one of the two pictures during each trial. Following each response, the cards were removed from the table and the therapist responded by saying “thank you” in an effort to avoid any training during this phase. After each trial, the position of the cards was counterbalanced quasi randomly and the cards were presented to the participant to start the next trial. If the child incorrectly identified a card for 60% or more of 5 presentations, the card was considered to be unknown and was included in the subsequent phases of the study. The pre-treatment phase was repeated until four unknown picture cards were identified. Once four cards were identified, the participant moved into one of the two training conditions. Thus, the pre-treatment assessment occurred multiple times throughout the study, at the beginning of each teaching condition, as to ensure that the stimuli were unknown just prior to the training procedure.

Table 1
Target stimuli taught for each training set for each participant.

Participant	Training Set 1	Training Set 2	Training Set 3	Training Set 4	Training Set 5
Brad	Circle, Arrow	Crying, Smiling	Tail, Fly	Write, Cut	Full, Closed
Julie	Red, Green	Tail, Feathers	Swim, Egg	Trunk, Wheels	
Sal	Sip, Pour	Sew, Dig	Feather, Shell	Horn, Mane	

1.6. Receptive discrimination training of unknown cards

Once four cards were identified as unknown, participants were randomly assigned to one of two discrimination training procedures. For both discrimination training procedures, two of the four unknown cards were randomly identified as “target” cards, or cards to be taught during discrimination training, and the other two were identified as “distracter” cards, and were not specifically taught. Participants received massed trials of each targeted picture card until reaching criteria (80% correct or above for 2 consecutive sessions). All sessions consisted of 10 trials.

1.6.1. Simple/conditional procedure

During this training procedure, discrimination training progressed from a simple conditional (no distracter) training phase to a conditional discrimination (distracter present) training phase. The purpose of this phase was to determine the number of sessions required to learn a discrimination using a “no distracter procedure.” Training began by presenting one of the two target cards in isolation (without a distracter present), and asking the child to “point to [target].” Correct responses were defined as the participant pointing to card the therapist had named and prompted responses were defined as the participant being physically guided to make the correct response. Responses were followed by praise and access to a toy for approximately 30 s or a small amount of an edible item on a variable ratio schedule of reinforcement (VR3). After each trial, the position of the card was counterbalanced quasi randomly in an array of one, and the card was presented to the participant to start the next trial. Sessions continued until the participant reached mastery in identifying the presented target card.

Next, the target card was presented along-side one of the unknown distracter cards for each trial. After each trial, the positions of the cards were counterbalanced quasi randomly in an array of two and the cards were presented to the participant to start the next trial. Here, the participant was required to identify the target card until reaching mastery. Once the first target card was considered mastered, the procedure was repeated with the second target card. Again, the second target card was initially presented in isolation (no distracter) and then along-side the second unknown distracter card. The participant was required to meet mastery criteria (80% for 2 consecutive sessions) for both the simple discrimination (no distracter) and the conditional discrimination (distracter present) before each targeted card was considered mastered.

1.6.2. Conditional only procedure

In the second training procedure, discrimination training occurred exclusively within the context of conditional discriminations (distracter present), thus eliminating the initial (no distracter) step. The purpose of this phase was to determine the number of sessions required to learn a discrimination using a procedure in which the distracter was present at the start of training. Training began by presenting one of the identified target cards along-side an identified distracter card and requiring that the participant “point to [target].” Again, the position of the cards was counterbalanced quasi randomly in an array of two, and the cards were presented to the participant to start the next trial. Once the participant demonstrated mastery of the first target card, the second target card was presented along-side the second distracter card until mastery was achieved.

1.7. Conditional discrimination training of taught picture cards

Once each participant completed both teaching procedures and two target cards were taught to mastery using each procedure, the conditional discrimination training phase was introduced. The purpose of this phase was to determine the number of sessions required to teach a conditional discrimination using two picture cards. Because each set of two cards had previously been taught using one of the two previously described procedures, this phase would allow a comparison of training time required when using these two procedures. During this phase, two target cards taught to mastery in each procedure were subsequently paired (each serving as a distracter for the other) for additional discrimination training. During this phase of the study participants were asked to point to one of the two cards. On any given trial, the correct target card alternated in a random order. Correct responses

were defined as the participant pointing to the card the therapist had named and prompted responses were defined as the participant being physically guided to make the correct response. Responses were followed by praise and access to a toy for approximately 30 s or a small amount of an edible item on a variable ratio schedule of reinforcement (VR3). After each trial, the position of the card was counterbalanced quasi randomly in an array of one and the card was presented to the participant to start the next trial. Sessions continued until the participant reached mastery in identifying the presented target card. The participant was required to achieve a response rate of 80% correct for 2 consecutive sessions before the cards were considered mastered.

1.8. Maintenance probe

Maintenance probes were conducted 1-month following the completion of the discrimination training of paired taught picture cards. Maintenance probes were conducted in the same setting as the training trials. The purpose of this phase was to assess the maintenance across time of discriminations taught using both procedures. This phase would indicate if one training procedure would result in better maintenance across time than another procedure. The procedures used during the maintenance probes were identical to those used in the conditional discrimination training except that no consequences were provided for responses and only 1 session (10 trials) was conducted.

2. Results

Fig. 1 represents the number of sessions required to master untrained stimuli using both teaching procedures: (1) the procedure that progresses from a simple (no distracter) training phase to the conditional (distracter) training phase (labeled Simple/Conditional), or (2) the procedure that provided training exclusively within the context of conditional discriminations (labeled Conditional Only). During the initial training, when target cards were paired with non-target distracter cards, all three participants required more training sessions using the procedure that included a no distracter training phase. Overall, this appears to be a result of the requirement of an additional procedural step rather than slowed acquisition occurring as a result of the no distracter phase. In other words, participants generally required a similar number of trials to mastery once conditional discrimination training began, but the incorporation of an additional training procedure lengthened the training process.

Fig. 2 represents the number of sessions required to master conditional receptive discriminations when the two previously taught target cards were presented along side one another. When making conditional receptive discriminations of trained stimuli, Brad performed similarly for stimuli taught with either teaching procedure. For Julie and Sal, the number of sessions required to master conditional discriminations of trained stimuli varied. For Julie, two training sets (Set 1 and Set 5) required more training sessions to mastery (6 and 15 sessions respectively) when the initial teaching procedure did not include an initial no distracter training phase. For Sal, one training set (Set 3) required more training sessions to mastery (12 sessions respectively) when the initial teaching procedure did not include an initial no distracter training phase.

Fig. 3 represents conditional discriminations at a 1-month follow-up to assess for maintenance of the learned skills. All three participants maintained conditional receptive discriminations (ranging from 80 to 100% accuracy) at a 1-month follow-up, regardless of initial teaching procedure, with the exception of one training set for Julie (Set 5). Julie performed at 50% correct in the maintenance condition for Set 5.

3. Discussion

In this study, three children were taught receptive discriminations using two commonly used discrete trial teaching procedures. All three children generally required more initial training sessions when the procedure incorporated the use of a no distracter training step. This finding is somewhat expected in that the inclusion of the no distracter procedure incorporates an additional teaching step, thus procedurally requiring additional sessions to reach mastery. However, it is relevant to note that at

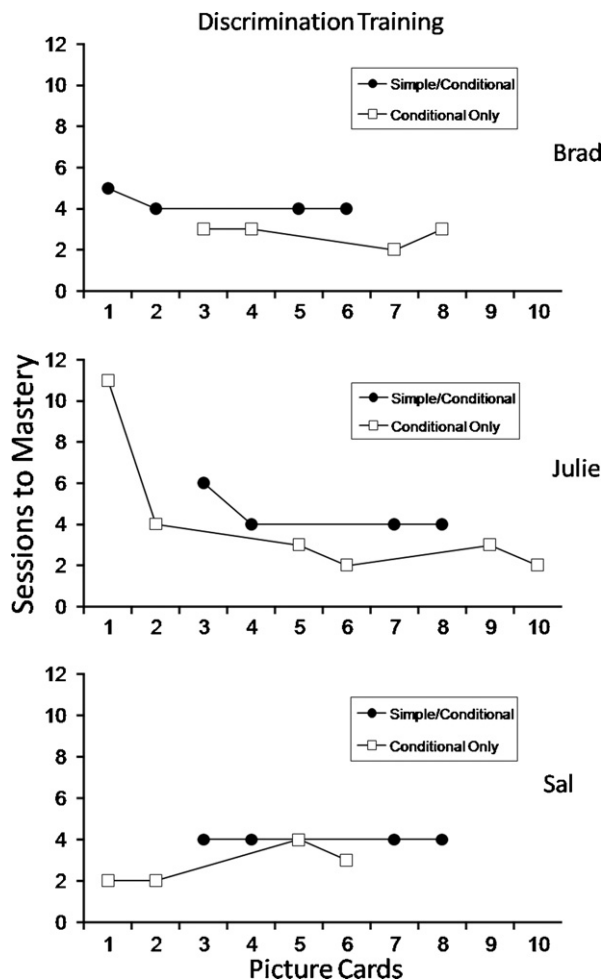


Fig. 1. Number of sessions required to master untrained stimuli using the simple/conditional and conditional only procedures.

least during the initial training, the inclusion of the no distracter training phase did not enhance the acquisition of receptive discriminations.

With regards to conditional discrimination of paired, previously taught, picture cards, the inclusion of the no distracter training procedure did not reliably facilitate acquisition of conditional discrimination of trained stimuli, although for Julie and Sal it was related to better performance in some instances. Brad acquired conditional receptive discriminations of trained stimuli quickly, regardless of teaching procedure. For Julie (Set 1 and Set 5) and Sal (Set 3), variability in data suggest that at least in some instances, receptive conditional discriminations of learned stimuli were mastered more quickly when they had been initially taught with a procedure that included the no distracter training procedure.

From a clinical standpoint, the inclusion of the no distracter training procedure exists in the absence of any empirical support. However, its use is common (Carr et al., 2006). We argue that for some children (as is the case with Brad), the inclusion of the no distracter training phase may have the disadvantage of allocating time to unnecessary practice trials and thus allocating time away from other instructional opportunities. Conversely, there may be instances when the inclusion of the no distracter training phase enhances acquisition of conditional discriminations. This appears to be the

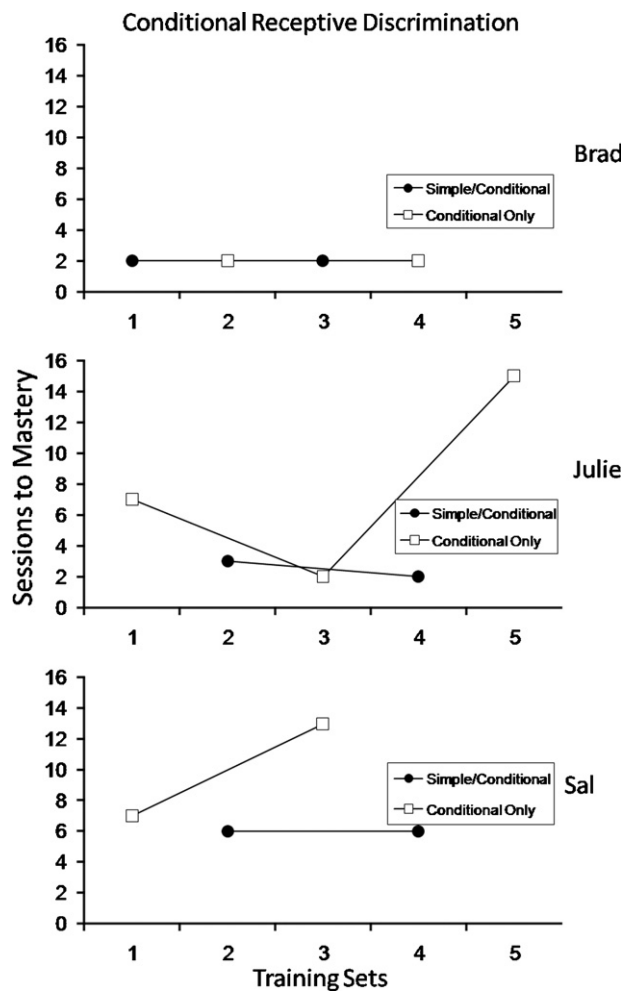


Fig. 2. Number of sessions required to master conditional receptive discriminations.

case for Julie and Sal in some instances. Additional research is recommended in order to help identify instances when this procedure may be beneficial. In addition, it may be useful clinically to assess the procedural utility on an individual case-by-case basis (using procedures similar to those described in this study). Ultimately, for individuals for whom there is no benefit to including the no distracter training phase, it could easily be abandoned clinically. Finally, there is no evidence to suggest that the inclusion of the no distracter training phase has a deleterious effect on acquisition of receptive discriminations. This is contrary to evidence presented by Terrace (1963) which suggested that discrimination training was enhanced when a distracter was presented earlier, rather than later, in training.

It is additionally notable that while this study presents an initial comparison of two commonly utilized discrete trial procedures for teaching receptive discriminations, additional alternative procedures have been proposed. For example, Green (2001) summarized techniques and recommended that receptive discriminations be taught within the context of multiple distracters, not just one as was done in this study. Thus, additional questions remain regarding the relative effectiveness of alternative discrimination teaching procedures.

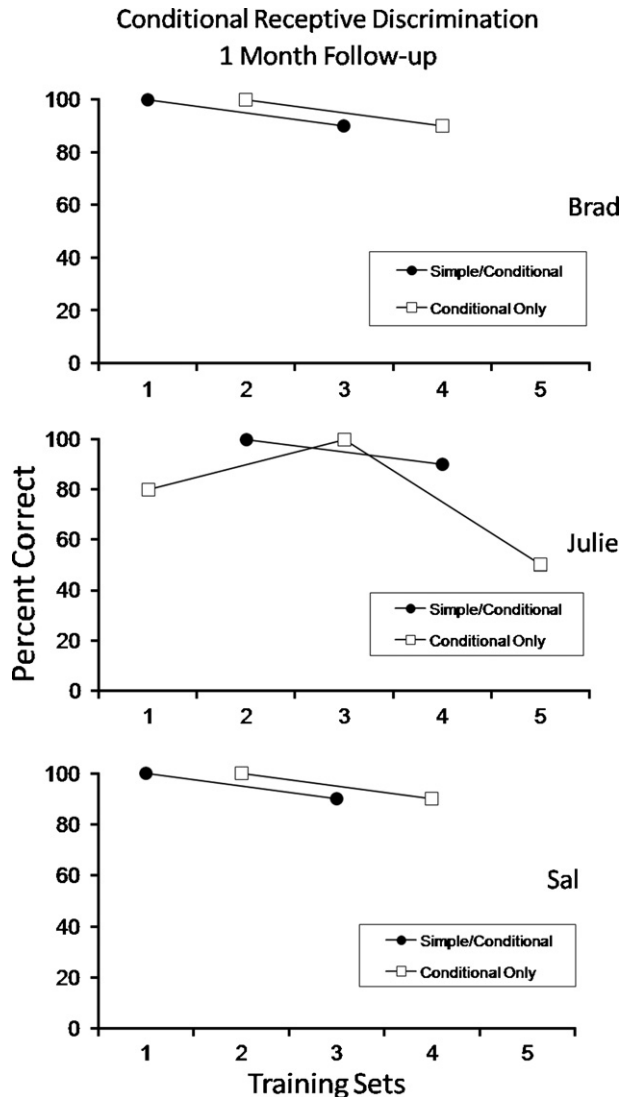


Fig. 3. Percent correct responses during conditional receptive discriminations taught using the simple/conditional and conditional only procedures at a 1-month follow-up.

Finally, while the results of this study provide preliminary evidence, additional research is required to assess if these findings will generalize to other individuals. For one, the three participants seemed to be relatively “quick learners.” In other words, for most of the items taught, only a few sessions were required for skill mastery. The learning patterns of certain children (e.g., quick learners) may be fundamentally different from children with different characteristics. Thus, it would be important to examine the relative effectiveness of both teaching procedures with children of different abilities. Second, while this study suggests that there may be little difference between the two training procedures, it may be important to evaluate the two procedures in a variety of contexts and with a variety of different skills. For example, it is possible that one procedure would be advantageous over another at teaching conditional discriminations of item features, classes of items, or item function. It will be necessary to conduct more thorough comparisons of the two procedures before a

determination can be made regarding the most effective and efficient method to teach discriminations using discrete trials.

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